

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **6524**

L61472986_L1505_48167

AY:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	202	Seat No:	2022600014
		Course Section:	1		



Name of the candidate. Harsh Gasula

Candidate's UID number:

2	0	2	2	6	0	0	0	1	4
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Examination class room number:

2	0	2
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Invigilator's signature with date:

	08/10/2025
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(To be filled in by the candidate)

EXAMINATION: BTECH MSE
(For example FY MCA/FY BTECH)

SEMESTER : VII

PROGRAM: CSE-AIMC

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: Wednesday 8/10/25

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Q.1a

According to the gradient descent algorithm
~~the~~ the weights are updated as follows:

$$w_i \leftarrow w_i + \Delta w_i$$

$$\text{where } \Delta w_i = -\eta \nabla E(w)$$

where η = learning rate
and $\nabla E(w)$ = gradient of the loss func.
w.r.t weights.

As we know,

$$\nabla E(w) = \left(\frac{dE}{dw_1}, \frac{dE}{dw_2}, \frac{dE}{dw_3}, \dots, \frac{dE}{dw_n} \right)$$

$$\text{Given that } E(w) = f(w) = \frac{1}{2} \|Xw - y\|^2$$

$$\therefore \nabla E(w) = 2 \times \frac{1}{2} (Xw - y)(X)$$

$$\nabla E(w) = (Xw - y)(X)$$

$$\therefore \Delta w_i = -\cancel{\eta} - \eta (Xw - y)(X)$$

$$\therefore \Delta w_i = \eta (y - Xw)(X)$$

$\therefore$ The gradient update rule becomes

$$w_i \leftarrow w_i + \eta (y_i - xw_i)(x)$$

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Q.1b

Bias → Occurs when the model capacity is low for the given data set, high bias leads to underfitting.

Variance → Occurs when the model capacity is high for the given dataset, high variance leads to overfitting.

Overfitting usually occurs when the model capacity is too high, this is because the model is designed for capturing more complex patterns than those present in the dataset when the patterns in the data are too simple or there isn't enough data for the model to be trained, overfitting occurs. ~~Overfit~~ Overfitting leads to poor generalisation as the model memorises the existing patterns in the data and fails to learn some of the complex patterns that it can use to generalise on unseen data. This way when the model overfits it typically performs well on the training data but performs poorly on the test data.

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Q.2a

When we apply a conv. filter over an input image, the output size is for a filter: $F \times F$

input dims $\rightarrow H \times W$

output dims $\rightarrow H_0 \times W_0$

$$H_0 = \frac{H - F}{s} + p \quad W_0 = \frac{W - F}{s} + p$$

where s = strides and p = padding
For K filters, the output channel size is K .

For every pixel in the output image, we perform $F \times F$ multiplications.

$\therefore$ The total number of multiplications,

$$= (F \times F) \times H_0 \times W_0 \times K$$

$$= (F \times F) \times \left(\frac{H - F}{s} + p \right) \times \left(\frac{W - F}{s} + p \right) \times K$$

~~Depthwise~~ Depthwise separable convolutions are good in terms of computational complexity as they involve lesser number of multiplications as compared to a standard conv operation. Hence, they are widely used in modern architectures to save on computational cost and parameter efficiency.

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d.3a

As we know for an RNN,

$$h_{t+1} = \sigma(W_{xh} x_{t+1} + W_{hh} h_t + b_h)$$

$$y_{t+1} = \tanh(W_{hy} h_{t+1} + b_y)$$

By chain rule of derivation:

$$\frac{dL}{dh_{t-k}} = \frac{dL}{dy_{t+k}} \times \frac{dy_{t+k}}{dh_{t-k}}$$

As we go deeper into the network, the equation of gradient descent becomes a high number of entities which might involve be <1 or >1 , which lead to vanishing or exploding gradients.

The forget gate in an LSTM controls how much of the past input should be used to update the memory cell. This way it retains all the required previous information even if the network becomes deep. The memory cell update eqⁿ look like:

$$c_{t+1} = f_t \odot c_t + i_t \odot \tilde{c}_t$$

Due to the ~~plus~~ 'plus' operation in the weight operation, when you take the derivative for updation they don't form a recursive long chain of multiplications which happened in case of an RNN and the gradients can be updated in a stable manner.

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Q.3b

The GRU involves two gates the update gate u and the reset gate r .

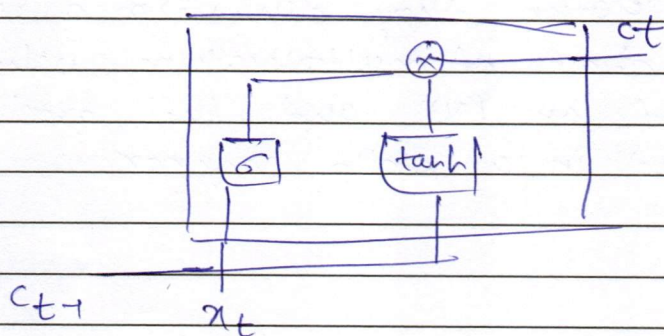
$$r = \sigma(w_r [x_t, c_{t-1}] + b_r)$$

$$u = \sigma(w_u [x_t, c_{t-1}] + b_u)$$

$$\tilde{c}_t = \tanh(r \cdot w_c [x_t, c_{t-1}] + b_c)$$

$$c_t = u \cdot \tilde{c}_t + (1-u)(c_{t-1})$$

GRU's only involve two gates whereas an LSTM involves three gates and a memory cell. It requires two lesser weight matrices as compared to a GRU. LSTMs are hence more computationally expensive as compared to a GRU and are of a high model capacity. For simple sequence based classification tasks involving a 1D signal, LSTMs might overfit due to high model capacity and, in such scenarios GRU's might be able to generalise on the input data better and hence lead to a better performance. For example: an audio classification task involving a 1D signal.



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